Squares and Power Sequences in the Reversible Square-Multiplication Graph

UNNS-U2-SQ – version 1.0 (v001)

UNNS Research Collective

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Abstract

Let G be the graph on the positive integers in which kN is joined to $k(N+1)^2$ for all $k, N \geq 1$, and let d be its graph distance. We prove that consecutive squares are at distance at most 5 and that 5 is attained: $\sup_{n\geq 1} \mathrm{d}(n^2, (n+1)^2) = 5$. The upper bound comes from 24 explicit path identities, one for each residue class of n modulo 24; in each class the path from n^2 to $(n+1)^2$ has at most five edges, its labels are polynomials in a class parameter, and every step is certified by an integral positive cofactor. The lower bound is the disjointness of the radius-two balls around $23^2 = 529$ and $24^2 = 576$. More generally, an edge $x-y$ lifts to a chain $x^k, x^{k-1}y, \ldots, y^k$ of k edges, so $\mathrm{d}(x^k, y^k) \leq k\,\mathrm{d}(x, y)$ and $\mathrm{d}(n^k, (n+1)^k) \leq 6k$. The path identities have strictly increasing polynomial intermediate states, so the parameter can be read back from the current integer; this yields deterministic finite-control selectors with no external input that pass through $1^2, 2^2, 3^2, \ldots$ with at most five moves per square, and through $1^k, 2^k, 3^k, \ldots$ with at most $6k$ moves per power.

1 Introduction

Provenance. The graph G is the undirected closure of the SELECT-II transformation $m \mapsto m(N+1)^2/N$ $(N \mid m)$ of the UNNS Substrate. Its uniform successor bound $\mathrm{d}(n, n+1) \leq 6$ is established in UNNS-U1-BND [3], and the finite-control successor selector (43 control states) in UNNS-U2-001 [4]. Both releases, the earlier notes [1, 2] and the frozen reference collection [5, 6] are left unchanged. The present release is the separate UNNS-U2-SQ baseline for squares and power sequences.

Question. Powers n^k are the simplest algebraic progressions after the integers themselves. The successor selector of [4] gives a route from $n-1$ to n in at most six moves; how far apart are n^2 and $(n+1)^2$ really, and can a fixed rule with no external input choose the path?

Theorem 1.1 (Summary)**.** *(a)* $\mathrm{d}(n^2, (n+1)^2) \leq 5$ *for all* $n \geq 1$*, and* $\mathrm{d}(23^2, 24^2) = 5$*; hence* $\sup_n \mathrm{d}(n^2, (n+1)^2) = 5$*.*

(b) For adjacent or arbitrary x, y *and* $k \geq 1$*,* $\mathrm{d}(x^k, y^k) \leq k\,\mathrm{d}(x, y)$*; in particular* $\mathrm{d}(n^k, (n+1)^k) \leq 6k$*. Also* $\mathrm{d}(n^2, (n+1)^2) \leq 8$ *by a short argument not using (a).*

(c) There is a deterministic selector with 85 *control states, reading only the current integer, that returns to its ready state exactly at* $1^2, 2^2, 3^2, \ldots$ *with at most five moves between consecutive squares. There is also a* 45*-state selector doing the same in at most eight moves, built on the successor selector of [4]. For each fixed* k *a finite-control selector realizes* $1^k, 2^k, 3^k, \ldots$ *in at most* $6k$ *moves per step.*

2 Preliminaries

A move $+N$ from x requires $N \mid x$ and gives $x(N+1)^2/N$; a move $-N$ requires $(N+1)^2 \mid x$ and gives $xN/(N+1)^2$. These are mutually inverse and the edges of G are exactly the pairs they join.

Lemma 2.1 (Scaling)**.** *If $\{x, y\}$ is an edge, so is $\{mx, my\}$ for every $m \geq 1$. Hence $\mathrm{d}(ma, mb) \leq \mathrm{d}(a, b)$.*

Proof. Multiply the cofactor k in $x = kN$, $y = k(N+1)^2$ by m. □

Lemma 2.2 (Power lifting)**.** *For $k \geq 1$ and any x, y: $\mathrm{d}(x^k, y^k) \leq k\,\mathrm{d}(x, y)$. For an edge $\{x, y\}$ the chain x^k, $x^{k-1}y$, …, y^k is a path of k edges in which every step carries the label of the original edge.*

Proof. Consecutive terms are $x^{k-j}y^j = m\,x$ and $x^{k-j-1}y^{j+1} = m\,y$ with $m = x^{k-j-1}y^j$, an edge by Lemma 2.1; write $x = cN$, $y = c(N+1)^2$ and the move is $+N$ with cofactor mc. Apply this along a shortest path from x to y. □

Corollary 2.3. $\mathrm{d}(n^k, (n+1)^k) \leq 6k$*; in particular* $\mathrm{d}(n^2, (n+1)^2) \leq 12$*, by* $\mathrm{d}(n, n+1) \leq 6$ *[3].*

Lemma 2.4 (Square-specific route)**.** *For $n \geq 2$, $n^2 \xrightarrow{-(n-1)} n-1$ and $n \xrightarrow{+n} (n+1)^2$ are legal moves. Hence $\mathrm{d}(n^2, (n+1)^2) \leq 1 + \mathrm{d}(n-1, n) + 1 \leq 8$.*

Proof. $(n-1+1)^2 = n^2$ divides n^2 and $n^2(n-1)/n^2 = n-1$; and $n \mid n$ with $n(n+1)^2/n = (n+1)^2$. For $n = 1$, $1 \to 4$ is an edge. □

3 The exact constant

Theorem 3.1. $\sup_{n \geq 1} \mathrm{d}(n^2, (n+1)^2) = 5$.

3.1 Upper bound: 24 path identities

Write $n = 24t + r$ with $0 \leq r < 24$. For each r Appendix A gives at most five signed labels $N_1(t), \ldots, N_\ell(t)$, $\ell \leq 5$, polynomials in t.

Lemma 3.2 (Certificates)**.** *For each r and all integers $t \geq t_0(r)$ the labelled path starting at n^2 is legal at every step and ends at $(n+1)^2$.*

Proof. Compose the moves symbolically. At each step the current state equals $c \cdot N$ (forward) or $c\,(N+1)^2$ (inverse) with a cofactor $c(t)$; the verifier checks by exact polynomial division that $c(t)$ is a polynomial, integer valued, and at least 1 for $t \geq t_0$ (all coefficients nonnegative after shifting $t \mapsto t_0 + u$ and value ≥ 1 at $u = 0$), and that the labels are integer valued and positive; the final state is $(n+1)^2$ as a polynomial identity. All 24 certificates pass (check Q1 of Table 1). □

Remark 3.3 (Example)**.** For $n = 24t + 1$ the states are

$$(24t{+}1)^2 \xrightarrow{+(24t+1)^2} 4(288t^2{+}24t{+}1)^2 \xrightarrow{-24t(12t+1)} 96t(12t{+}1)$$
$$\xrightarrow{+6t} 16(6t{+}1)^2(12t{+}1) \xrightarrow{-(12t+1)} 4(12t{+}1)^2 = (n+1)^2.$$

Only $n \in \{1, 2, 3\}$ lie below t_0 in their classes; for these the paths $1 \to 4$, $4 \to 16 \to 100 \to 9$ and $9 \to 100 \to 16$ are listed in the data file and verified. Hence $\mathrm{d}(n^2, (n+1)^2) \leq 5$ for all n.

3.2 Lower bound

Lemma 3.4. $\mathrm{d}(23^2, 24^2) = 5$.

Proof. The radius-two balls $B_2(529)$ and $B_2(576)$ have 97 and 562 vertices and are disjoint, computed by two independent neighbour-enumeration programs; so no path of length ≤ 4 exists. The path $529 \to 280900 \to 1056 \to 13824 \to 3072 \to 576$ with moves $+529, -264, +11, -2, -3$ has length five. □

This proves Theorem 3.1. The lower bound is a computation; the exact distance table for $n \leq 500$ shows the value 5 for 94 values of n, the first being 23, 28, 36, 40, 43, 47. Whether 5 occurs for infinitely many n is open.

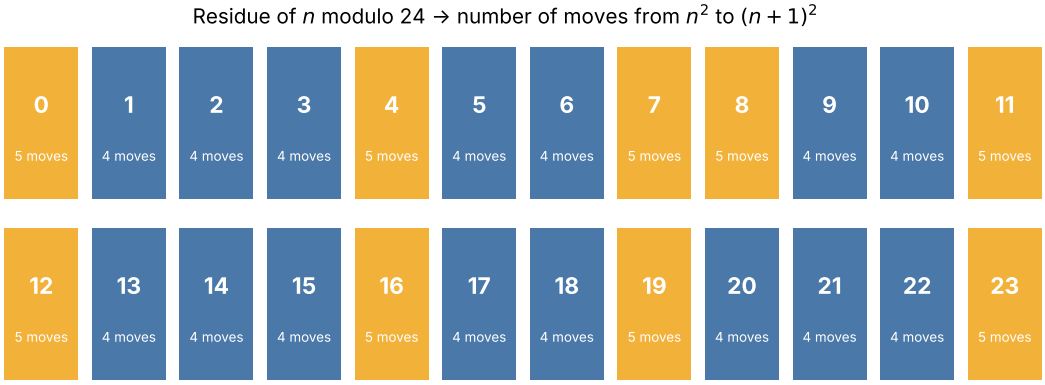


Figure 1: The 24 residue classes and the number of moves of their identity (at most five).

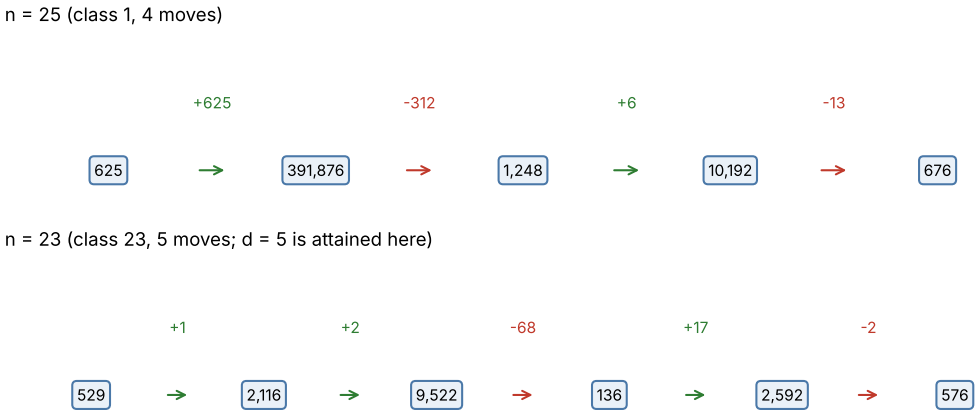


Figure 2: Two paths from n^2 to $(n+1)^2$ generated by the identities (green: forward move $+N$, red: inverse move $-N$).

distance $d(n^2, (n+1)^2)$	number of $n \leq 500$
1	1
2	2
3	61
4	342
5	94

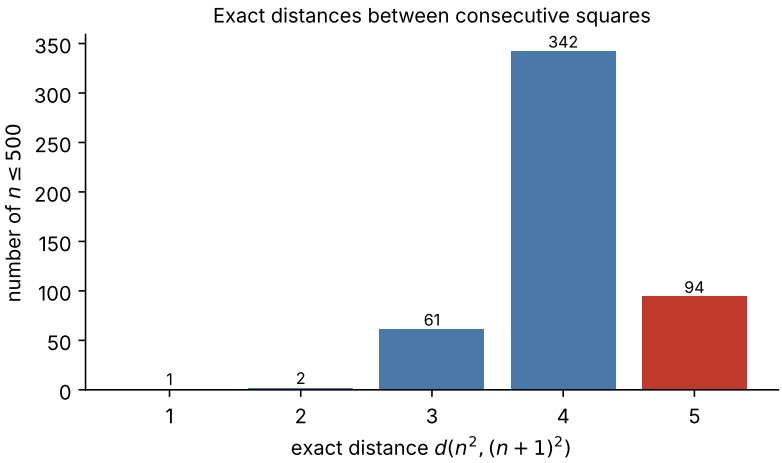


Figure 3: Exact distances between consecutive squares for $n \leq 500$ (mean 4.052); the identities do not always give the shortest path but never exceed five moves.

4 Finite-control generation of squares and powers

We use the selector model of [4]: a finite set of control states, a distinguished ready state, and a move rule reading exact arithmetic on the current integer and the control state only; no target, counter or other external datum.

Lemma 4.1 (Recoverability)**.** *In each of the 24 identities the states $s_0(t) = n^2, s_1(t), \ldots, s_\ell(t) = (n+1)^2$ are strictly increasing polynomials of $t \geq t_0$. Thus t is determined by the template, the phase and the current integer.*

(Symbolic derivative test on the shifted polynomials; check Q7.)

Theorem 4.2 (Direct square selector)**.** *The selector Σ_{sq} with 85 control states (the ready state, the template phases of the 24 classes, and the phases of the three exceptional paths) is deterministic, uses only (x, q), and from $(1, \mathsf{READY})$ returns to READY exactly at $2^2, 3^2, 4^2, \ldots$ in order, with at most five moves between consecutive ready states.*

Proof. At READY with $x = n^2$ take $n = \lfloor \sqrt{x} \rfloor$, read $r = n \bmod 24$ and $t = (n - r)/24$, and play the first label. At phase i recover t from $s_i(t) = x$ (Lemma 4.1) and play $N_{i+1}(t)$. By Lemma 3.2 every move is legal and the last state is $(n+1)^2$. □

Theorem 4.3 (Wrapper selector)**.** *The 45-state selector below is deterministic and returns to $\mathsf{SQ_READY}$ at $1^2, 2^2, 3^2, \ldots$ in at most eight moves per step. At a square $x = n^2$ with $n \geq 2$ it plays $-(n-1)$ to reach $n - 1$, runs the successor selector of [4] (each of its 43 control states wrapped), reaches n, and plays $+n$ to reach $(n+1)^2$ (Lemma 2.4); at $x = 1$ it plays $+1$.*

The wrapper reuses the frozen successor selector without change; it needs $43 + 2 = 45$ control states.

Theorem 4.4 (Fixed powers)**.** *For each $k \geq 1$ the successor identities of [3], lifted by Lemma 2.2 (each move repeated k times with intermediate states $s_i^{k-j} s_{i+1}^{j}$, again strictly increasing in t), give a finite-control selector that returns to its ready state at $1^k, 2^k, 3^k, \ldots$ with at most $6k$ moves per step. The control depends on k.*

exponent k	bound $6k$	max moves observed	control states
2	12	12	96
3	18	18	149
4	24	24	202
5	30	30	255

Remark 4.5 (Scope)**.** The exponent k is fixed in advance; no single controller for all k is claimed. Finite control is shown to be sufficient for squares, not necessary: unlike the identity sequence, squares leave non-square integers available as intermediate states, so the memoryless obstruction of [4] does not transfer, and whether a state-only square generator exists is open.

5 Validation, status and open questions

Over 20000 successive squares the direct selector made at most five moves per step with mean 4.3747; the wrapper's mean over 5000 squares was 6.915. Every move was checked against the edge definition.

statement	status	basis
$\sup d(n^2,(n+1)^2) = 5$, upper bound	proved	24 certificates (Q1); replay to 3×10^5 (Q2)
same, lower bound	computation	$d(23^2,24^2) = 5$, two engines (Q3)
exact table	computation	$n \le 500$ (Q4), second engine $n \le 40$ (Q5)
power lifting, $6k$ bound	proved	Lemma 2.2; checked (Q6)
direct and wrapper selectors	proved, computer-supported	Q7–Q10
fixed-power selectors	proved, computer-supported	$k \le 5$ run (Q11)
formal proof-assistant certification	not done	–
literature originality review	not done	–

Table 1: Status. All checks passed (337995 cases; verifier hash prefix `d2f57b990c511d7f`).

Open questions.

1. Does $\mathrm{d}(n^2,(n+1)^2) = 5$ occur infinitely often? Is there a residue-class structure beyond modulus 24 that lowers the identity length?
2. Fibonacci numbers and primes: which uniform distance bounds hold between consecutive terms, and what replaces the recoverable-parameter property of the path identities? For primes the unbounded search suggests finite control alone does not suffice.
3. Is there a single finite controller realizing all exponents k uniformly, or a state-only square generator?

A The 24 identities

Each row lists the signed labels $N_1(t),\dots,N_\ell(t)$ of the path from n^2 to $(n+1)^2$ for $n = 24t+r$ (a move $+N$ is $x \mapsto x(N+1)^2/N$, a move $-N$ is $x \mapsto xN/(N+1)^2$).

r	$t \geq$	moves (labels $N_i(t)$ for $n = 24t + r$)
0	1	$+12t,\ +12t,\ +1,\ -(24t+1),\ -(24t+1)$
1	1	$+(24t+1)^2,\ -24t\,(12t+1),\ +6t,\ -(12t+1)$
2	1	$-12t,\ +6t,\ -(12t+1),\ +2\,(12t+1)$
3	1	$+3,\ -1,\ -8t,\ +6t$
4	0	$+1,\ +2\,(6t+1),\ +2\,(6t+1),\ -(24t+5),\ -(24t+5)$
5	0	$+(24t+5)^2,\ -12\,(4t+1)\,(6t+1),\ +(6t+1),\ -3\,(4t+1)$
6	0	$-2\,(6t+1),\ +(6t+1),\ -3\,(4t+1),\ +6\,(4t+1)$
7	0	$+1,\ +2,\ -4\,(18t+5),\ +(18t+5),\ -2$
8	0	$+1,\ +1,\ -3\,(16t+5),\ +(16t+5),\ -3$
9	0	$+9\,(8t+3)^2,\ -8\,(3t+1)\,(12t+5),\ +2\,(3t+1),\ -(12t+5)$
10	0	$-4\,(3t+1),\ +2\,(3t+1),\ -(12t+5),\ +2\,(12t+5)$
11	0	$+1,\ +2,\ -8\,(9t+4),\ +2\,(9t+4),\ -2$
12	0	$+6\,(2t+1),\ +6\,(2t+1),\ +1,\ -(24t+13),\ -(24t+13)$
13	0	$+(24t+13)^2,\ -12\,(2t+1)\,(12t+7),\ +3\,(2t+1),\ -(12t+7)$
14	0	$-6\,(2t+1),\ +3\,(2t+1),\ -(12t+7),\ +2\,(12t+7)$
15	0	$+3,\ -1,\ -4\,(2t+1),\ +3\,(2t+1)$
16	0	$+1,\ +4\,(3t+2),\ +4\,(3t+2),\ -(24t+17),\ -(24t+17)$
17	0	$+(24t+17)^2,\ -24\,(3t+2)\,(4t+3),\ +2\,(3t+2),\ -3\,(4t+3)$
18	0	$-4\,(3t+2),\ +2\,(3t+2),\ -3\,(4t+3),\ +6\,(4t+3)$
19	0	$+1,\ +2,\ -8\,(9t+7),\ +2\,(9t+7),\ -2$
20	0	$-2\,(3t+2),\ +(3t+2),\ -(4t+3),\ +2\,(4t+3)$
21	0	$+9\,(8t+7)^2,\ -4\,(6t+5)\,(12t+11),\ +(6t+5),\ -(12t+11)$
22	0	$-2\,(6t+5),\ +(6t+5),\ -(12t+11),\ +2\,(12t+11)$
23	0	$+1,\ +2,\ -4\,(18t+17),\ +(18t+17),\ -2$

B Reproducibility

From `code/`: `python3 verify_u2_sq.py --out ../reports/u2_sq_verification_report.json --csv ../data/square_distance_table.csv` (about nine minutes; run without `-I`). The files `u1_bnd_lib.py` and `u2_selector.py` are unchanged copies of the earlier releases. The identities were found by a search over tuples of states at test points (`sqparam.py`) and resolved and certified symbolically (`sqsym.py`); only the certified identities are used by the selectors. The heavy verification of the earlier releases is not re-run (see the README).

References

[1] UNNS Research Collective, *UNNS-U0: Arithmetic reachability and orbit admissibility*, research note U0-001, 8 November 2026 (local release; not part of the frozen repository).

[2] UNNS Research Collective, *UNNS-U1: Reversible arithmetic connectivity*, research note, 8 November 2026 (provisional baseline; local release).

[3] UNNS Research Collective, *Bounded Successor Distance in the Reversible Square-Multiplication Graph*, UNNS-U1-BND version 1.0 (v001), 8 November 2026 (local release).

[4] UNNS Research Collective, *Intrinsic Generative Selectors: Finite Control, Arithmetic State, and Observational Realization*, UNNS-U2-001 version 1.0 (v001), 8 November 2026 (local release).

[5] UNNS Research Collective, *UNNS Generative Information Geometry collection*, frozen release, repository `ukbbi/UNNS`, commit `a115e4bc6b8ee734e6146b3b17a6b121e65d31c3`: https://github.com/ukbbi/UNNS/tree/a115e4bc6b8ee734e6146b3b17a6b121e65d31c3/projects/UNNS_Generative_Information_Geometry.

[6] UNNS project repository, pinned at the same commit: https://github.com/ukbbi/UNNS/tree/a115e4bc6b8ee734e6146b3b17a6b121e65d31c3.